YEAR 7 · UNIT 1 · EXTRA HOMEWORK

Two Rows, Then Add

Two-digit multiplication · and the division that undoes it

Name: ______________________________ **Class:** ______________________________

Date given: ________________________ **Date due:** __________________________

Why this sheet exists

The list of square numbers you copied stops at 12×12. Every square after that is a **two-digit multiplication**: 13×13, 18×18, 24×24. And a cube is **two** of them — to check that $\sqrt[3]{5832} = 18$ you work out $18 \times 18 = 324$, and then 324×18.

So this is not practice **beside** Unit 1. It is the arithmetic Unit 1 **finishes on**, and the sheet ends on the exact divisions Homework 4 asks you for.

No calculator, and show your working — working you can see is working you can check.

Part 1 · The facts the method runs on

E1 Fill in the missing number

Two-digit multiplication is these facts, done twice and added. Aim to know them without counting.

(a) $6 \times 8 =$ ______	(f) $9 \times 7 =$ ______	(k) $8 \times 4 =$ ______
(b) $7 \times 8 =$ ______	(g) $6 \times 6 =$ ______	(l) $7 \times 7 =$ ______
(c) $9 \times 4 =$ ______	(h) $9 \times 8 =$ ______	(m) $12 \times 8 =$ ______
(d) $7 \times 6 =$ ______	(i) $7 \times 4 =$ ______	(n) $11 \times 7 =$ ______
(e) $8 \times 8 =$ ______	(j) $9 \times 9 =$ ______	(o) $12 \times 6 =$ ______

E2 Multiply by a ten

The second row is always a ten

To multiply by **20**, multiply by **2** and then **put a zero on the end**. $34 \times 2 = 68$, so $34 \times 20 = 680$. That is the whole trick, and it is what the second row of a two-digit multiplication is.

(a) $23 \times 10 =$ ________	(c) $41 \times 30 =$ ________	(e) $56 \times 20 =$ ________
(b) $23 \times 20 =$ ________	(d) $17 \times 40 =$ ________	(f) $38 \times 50 =$ ________

Part 2 · Two rows, then add

The method, once, in full

$$\begin{array}{r} 4\,7 \\ \times\quad 2\,6 \\ \hline 2\,8\,2 \\ 9\,4\,0 \\ \hline 1\,2\,2\,2 \\ \hline \end{array}$$

Row 1: $47 \times 6 = 282$.
Row 2: $47 \times 20 = 940$ — multiply by 2, then put the zero on.
Add them: $282 + 940 = 1222$.

If you only write one row, you have multiplied by 6 and forgotten the 20. Your answer will come out about ten times too small. **Two rows, every time.**

E3 Warm up: two digits by one digit

Set each one out properly and show your working in the box.

46×7	58×6	73×8	89×4

E4 Two digits by two digits

Two rows, then add. Show both rows every time.

23×14	34×16	27×25
58×17	39×41	75×36
46×32	64×23	52×48

84 × 26	47 × 53	96 × 15

Look back. Two of those twelve have the **same answer**. Which two? ____________________

E5 Squaring: the same number, twice

Where your list of squares runs out

Your notebook has the squares up to 12 × 12 = 144. After that there is no list — you **multiply**. 13^2 means 13 × 13, and that is a two-digit multiplication like any other.

Work each one out. **You will meet three of these answers again in Homework 4.**

13 × 13	15 × 15	16 × 16

18 × 18	24 × 24	25 × 25

Now use one. Your answer to 18 × 18 is the first half of a check in Homework 4. Finish it here — this one is **three digits by two**, so it takes two rows like all the rest.

18 × 18 = __________, and then __________ × 18 = _____________

Part 3 · Division, which undoes it

E6 Divide — these all come out exactly

A check that costs five seconds

When you finish a division, **multiply your answer back**. If 84 ÷ 6 = 14, then 14 × 6 must come to 84. If it does not, you have just caught your own mistake.

(a) $84 \div 6 =$ __________

(b) $92 \div 4 =$ __________

(c) $135 \div 5 =$ __________

(d) $168 \div 7 =$ __________

(e) $216 \div 8 =$ __________

(f) $288 \div 12 =$ __________

(g) $405 \div 9 =$ __________

(h) $728 \div 8 =$ __________

(i) $924 \div 7 =$ __________

(j) $576 \div 4 =$ __________

E7 Divide — these do not

Word help — the words that describe what is left

exactly	nothing is left over. $84 \div 6 = 14$ exactly.
remainder	what is left over when it does not come out exactly. We write $75 \div 8 = 9$ r 3.
goes into	"8 goes into 24 three times" means $24 \div 8 = 3$.
left over	the everyday English for a remainder.

Write each answer as a whole number and a remainder, like this: $75 \div 8 = 9$ r 3.

(a) $100 \div 6 =$ ____________

(b) $131 \div 9 =$ ____________

(c) $250 \div 7 =$ ____________

(d) $194 \div 12 =$ ____________

(e) $365 \div 4 =$ ____________

(f) $500 \div 8 =$ ____________

(g) $425 \div 6 =$ ____________

(h) $800 \div 9 =$ ____________

E8 Find the missing number

Multiplying and dividing undo each other

If $14 \times 13 = 182$, then $182 \div 14 = 13$ and $182 \div 13 = 14$. Every multiplication fact is also two division facts.

(a) $14 \times$ ________ $= 182$

(b) ________ $\times 16 = 384$

(c) $576 \div$ ________ $= 24$

(d) ________ $\div 9 = 49$

(e) $25 \times$ ________ $= 625$

(f) ________ $\times 18 = 324$

(g) $1296 \div$ ________ $= 144$

(h) ________ $\div 8 = 216$

E9 Divide by a two-digit number

Same method, bigger question

Dividing by 24 is the same as dividing by 6. You still ask **how many fit**, you are just asking it about a bigger number. It helps to write the first few out first: 24, 48, 72, 96, 120, . . .
And the check is the same: **multiply your answer back**.

These all come out exactly. Show your working in the box.

$384 \div 16$	$950 \div 25$
$672 \div 24$	$585 \div 13$

E10 Use it: the four divisions Homework 4 asks for

Straight out of Homework 4

In Homework 4 you split a big number into two pieces you know the root of. The **splitting is a division**, and these are the four it needs. Do them here and you have done them there.

The division	Answer	So the split is
$441 \div 9$		$441 = 9 \times$ ________
$1296 \div 9$		$1296 = 9 \times$ ________
$1728 \div 8$		$1728 = 8 \times$ ________
$5832 \div 8$		$5832 = 8 \times$ ________

E11 Word problems

Word help — which operation does the sentence want?

each / **every**	usually multiply. 24 boxes with 36 in **each** is 24×36.
share equally between	divide.
how many are left over	the remainder.
altogether / **in total**	the whole amount, once you have finished.

Write the calculation first, then the answer.

1. **The exercise books.** Mr Bowen orders **24 packs** of exercise books. There are **36 books** in each pack. How many books does he have altogether?

2. **The chairs.** There are **950 chairs** in the hall, set out in rows of **24**. How many full rows are there, and how many chairs are left over?

3. **The wall.** Mr Bowen tiles a wall with **26 rows** of **34 tiles**. How many tiles does the wall take altogether?

4. **The pencils.** Mr Bowen buys **27 boxes** of pencils with **36 pencils** in each box. He shares them equally between his **24 students**, and keeps whatever is left over for himself. How many pencils does each student get, and how many does Mr Bowen keep?